Beyond One Number: Controlled Pairwise Evaluation and Design Search for Camera Tracking

Kevin Blackburn-Matzen
Independent Researcher
https://github.com/kmatzen/plumbline

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Abstract

Camera tracking is usually compared with scalar pose metrics. These metrics are necessary but do not describe every failure that a person sees in a rendered trajectory. A camera-center metric can miss rotation corruption entirely, and an average can hide a short jump. We introduce Plumbline Review, an evaluation method built around reviewable artifacts and controlled pairwise decisions. It applies deterministic validity gates, renders each accepted trajectory with the same diagnostic views, collects blinded pairwise judgments using a fixed rubric, fits a Bradley–Terry preference model, and reports task-level bootstrap intervals. The resulting preference signal can also drive budgeted design search.

We validate the method in two controlled experiments. First, we inject drift, translation jitter, local jumps, and rotation jitter into 24 synthetic camera paths. The known defect order defines 240 pairwise trials. The full review score selects the less severe defect in 92.9% of trials, compared with 79.6% for absolute trajectory error alone. Second, we test the implemented Bayesian proposer on a mixed continuous, integer, and categorical search space. Across 60 trials with a budget of 20 evaluations, Bayesian search reaches within 0.05 of the known optimum in 48.3% of runs, compared with 10.0% for random search. These experiments establish controlled sensitivity and search behavior. They do not replace a human preference study on real videos, which is the main next validation step.

1 Introduction

A camera trajectory can have a good global alignment score and still look wrong. It may shake from frame to frame, rotate erratically, pause while the video moves, or fail for a short interval. Absolute trajectory error (ATE), relative pose error (RPE), and pose-threshold accuracy each reveal part of the result. None is a complete account of visible quality.

This creates a practical problem when choosing between camera-tracking methods. Researchers inspect videos, plots, point clouds, and per-frame errors, then make an informal judgment. The judgment may be useful, but it is hard to repeat. The views can differ between methods, labels may reveal the method, invalid outputs may disappear, and a final ranking may have no uncertainty estimate.

We propose Plumbline Review, a reproducible evaluation layer for camera tracking. It does not replace pose metrics. It turns the broader review process into a declared protocol. Every candidate first passes deterministic gates. Valid candidates become versioned artifacts containing poses, scalar metrics, and a standard diagnostic render. Candidate pairs are blinded and balanced. A reviewer answers the same concrete questions for each pair. Pairwise outcomes are aggregated with a statistical model, while the original metrics remain available for diagnosis.

This design draws on pairwise evaluation in 3DCodeBench, where automated measures alone do not capture all differences in generated 3D assets [2]. The camera problem requires added controls: trajectory conventions, Sim(3) alignment, non-finite poses, frozen tracks, and separation between automatic rejection and subjective review.

The contributions are:

1. an artifact schema and validity gate that make camera comparisons inspectable and resumable;
2. a controlled, blinded pairwise protocol with balanced assignment, Bradley–Terry aggregation, and task-level uncertainty;
3. a provider-neutral judge interface with a defined human-calibration test;
4. an uncertainty-aware Bayesian design loop for expensive tracking evaluations; and
5. controlled experiments that expose an ATE blind spot and measure search efficiency under a fixed budget.

2 Related Work

Camera metrics. Trajectory benchmarks commonly use global alignment followed by ATE, local relative errors, or pairwise rotation and translation thresholds. Recent

geometric models report several of these measures, including DUSt3R [6], MASt3R [3], MonST3R [9], CUT3R [7], VGGT [5], and Depth Anything 3 [8]. Multiple metrics help, but researchers still inspect outputs when the numbers disagree.

Reproducible evaluation. Clear protocols, reusable code, and retained artifacts improve reproducibility [4]. Evaluation harnesses in language modeling have shown that small implementation choices can alter conclusions [1]. Our work applies the same principle to camera review: the render, gate, pairing, rubric, and aggregation rule are part of the measurement.

Preference evaluation. 3DCodeBench combines automated measures with pairwise human preference for procedural 3D models [2]. We adopt pairwise comparison because it asks a narrower question than absolute scoring. Our method differs in its deterministic camera-validity stage, trajectory-specific artifacts, task-grouped uncertainty, and direct connection to design search.

3 Method

3.1 Review artifact

For system i on task t, the harness writes an immutable artifact

$$A_{it} = \{M, P, D, G, V\},$$

where M is a manifest, P is the predicted and reference pose data, D is a standard diagnostic image, G contains scalar metrics and per-frame summaries, and V records schema, renderer, code, and configuration versions. Dataset, sequence, system, and configuration identifiers are stored in the manifest. File hashes make later changes visible.

The diagnostic has fixed size and layout. It shows aligned paths from consistent viewpoints, per-frame normalized translation error, rotation error, and a metric panel. Pair images place two diagnostics side by side and use only labels A and B. A deterministic hash changes left-right order, so an interrupted run can resume without changing the assignment.

3.2 Validity before preference

Preference is not used to rescue malformed output. The gate checks pose-array shape, frame count, finite coverage, homogeneous rows, rotation orthogonality, rotation determinant, and predicted motion. A non-finite, malformed, or frozen trajectory is marked invalid before rendering. If one candidate is valid and the other is not, the gate records the outcome. If both are invalid, the pair is recorded as *both bad* and excluded from the preference fit. This keeps failure coverage visible without asking a reviewer to compare unreadable artifacts.

Valid trajectories are aligned to the reference with one global Sim(3) transform for diagnostic position errors. Rotation errors are computed separately. Alignment removes coordinate gauge but does not remove temporal jitter or a local failure.

3.3 Blinded pairwise review

For each task, the scheduler chooses the least-observed system pairs. Ties in global pair count are resolved lexically, which makes scheduling reproducible. The reviewer sees the pair image and applies a fixed rubric:

1. Does either trajectory contain a local jump or frozen interval?
2. Which trajectory has less frame-to-frame jitter?
3. Which rotation track is more stable and consistent with the reference?
4. Which trajectory better preserves the path over the full sequence?

The response is A, B, tie, or both bad. The log stores the comparison identifier, reviewer class, verdict, reason, artifact hashes, and time. System identities are joined only during aggregation.

An external vision-language judge can receive the same image and response schema. It is not treated as interchangeable with a person. Before use, its non-tie coverage, accuracy on shared decisive comparisons, confusion matrix, and Cohen's κ are measured against held-out human votes. The present paper tests the provider-neutral interface but does not report a human-calibrated model judge.

3.4 Preference model and uncertainty

For a decisive comparison between systems i and j, the Bradley–Terry model is

$$P(i \succ j) = \frac{\exp(q_i)}{\exp(q_i) + \exp(q_j)}.$$

A tie contributes one half-win to each system. Both-bad pairs do not enter the fit. A small L_2 term prevents numerical separation, and scores are centered because only differences are identifiable. The comparison graph must be connected.

Confidence intervals resample complete tasks rather than individual votes. This keeps all comparisons from one video together and avoids treating correlated views of the same scene as independent evidence. We report coverage and invalid-rate results next to preference scores.

3.5 Design search

The evaluator can be an expensive black-box objective for configuration search. Parameters may be continuous,

integer, or categorical. Completed observations contain the configuration, preference estimate or controlled quality score, runtime objectives, constraints, and status.

With fewer than three successful observations, the proposer uses maximin space filling. It then fits Gaussian-process surrogates with an RBF kernel. Multi-objective runs use a seeded random scalarization, following the ParEGO pattern, rather than one hidden permanent weighting. Expected improvement chooses the next candidate. Separate surrogates downweight candidates likely to violate constraints or fail. Every proposal stores its seed, objective weights, acquisition value, and configuration hash.

This loop does not make the test set part of training. Search uses a declared development set and budget; the chosen configuration is evaluated once on a held-out final set.

4 Controlled Experiments

The experiments answer two narrow questions. First, can the full evaluator distinguish trajectory defects that ATE alone misses? Second, does the Bayesian proposer use a small evaluation budget better than random sampling? All results are generated by `paper/run_experiments.py` with fixed seeds.

4.1 Trajectory-defect study

We generate 24 smooth, non-collinear camera paths with 80 frames. Each has translation and yaw motion. Five systems are formed from every path: clean, smooth translation drift, rotation jitter, translation jitter, and a short local jump. The perturbation families have a declared severity order: clean, drift, rotation jitter, translation jitter, local jump. This order represents known injected defects, not human opinion.

The full controlled review score combines the 95th percentile aligned translation error, mean acceleration magnitude, maximum jerk, and 95th percentile rotation error. Its weights and the defect order are fixed before comparison. The score is used only as a repeatable reviewer in this controlled test. All ten system pairs on all 24 tasks produce 240 trials. We compare its selections with selections made by lower ATE alone. We also fit the pairwise outcomes with the same Bradley–Terry implementation used by the review harness.

4.2 Design-search study

The search space contains a log-scaled continuous confidence threshold, an integer temporal window from 2 to 16, and a categorical precision choice. A smooth hidden function contains a narrow threshold optimum, a window optimum, an interaction term, and a small precision effect. Its maximum is computed on a dense grid. The optimizer sees only evaluated points.

Each of 60 trials starts from four shared random observations. Bayesian and random search then receive the same total budgets from 5 through 20. The primary measure is simple regret, the gap between the best value found and the grid maximum. We also report the fraction of trials within 0.05 of that maximum at budget 20.

5 Results

5.1 A broader review catches metric blind spots

Across 240 comparisons, the full review score selects the less severe injected defect in 92.9% of trials. ATE alone reaches 79.6%. Kendall's τ between the five-system mean ranking and the declared order is 0.80 for review and 0.74 for ATE.

The main blind spot is rotation. Rotation jitter has effectively zero ATE because camera centers do not change, yet its mean rotation RPE is 8.37 degrees. ATE therefore ranks the corrupted rotation track alongside the clean path. The full review includes rotation and places it below clean and drift. This result is not an argument to discard ATE. It shows that a position-only scalar cannot answer a broader quality question.

The full score does not perfectly recover the declared order. It places the local jump slightly above translation jitter in its fitted rating, leading to the 7.1% error rate. This is useful evidence: the result changes with the rubric's weighting of persistent roughness versus a short severe failure. A real study should expose those components and test the choice with people rather than hide it inside one score.

5.2 Bayesian search improves budget efficiency

At budget 20, median regret is 0.059 for Bayesian search and 0.236 for random search. Bayesian search reaches within 0.05 of the known optimum in 48.3% of trials; random search does so in 10.0%. Figure 1 shows the median regret path. The gain appears after the initial observations, when the Gaussian-process model begins to exploit the smooth response surface.

This experiment verifies the implementation on a mixed search space with a known optimum. It does not establish the same gain on a real tracking model. Real objectives may be noisier, less smooth, and more expensive, and preference uncertainty should then be passed into the search model.

6 Reproduction Harness as Supporting Evidence

The evaluator is implemented inside a broader protocol-first reproduction harness. Its canonical ledger currently

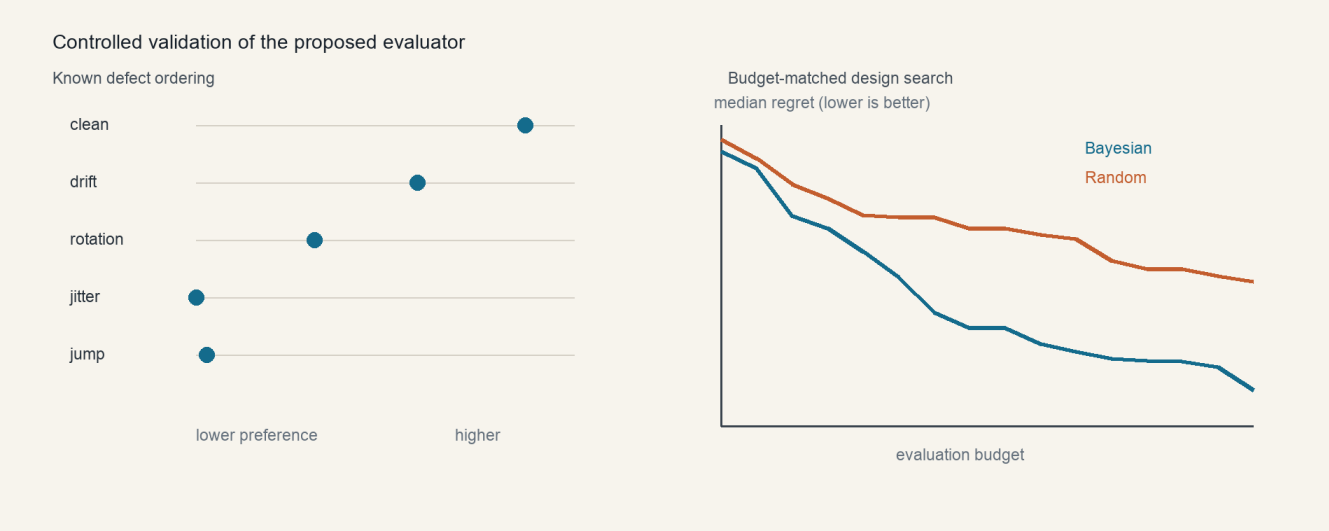


Figure 1: Controlled validation. Left: Bradley–Terry scores from the fixed review rule recover the broad known defect order; translation jitter and a local jump remain close. Right: Bayesian search reduces median regret faster than random search under the same budgets.

contains 47 verified published-result cells across 16 method variants and 13 datasets: 39 depth cells and 8 pose cells. The median absolute deviation from the paper value is 1.5%, the 90th percentile is 4.8%, and 46 of 47 cells are within 5%.

These numbers show that the infrastructure can preserve varied model, dataset, alignment, and metric protocols. They are not the main contribution and not a reproduction success rate. The ledger is curated, not randomly sampled, and excludes claims that have not reached a verified state.

7 Limitations and Next Validation

The trajectory study is controlled. Its severity order and review rule are designed, not collected from people. It establishes sensitivity to known defects and exposes a clear ATE blind spot, but it cannot show that the resulting ranking matches human preference on real videos. The highest-priority next experiment is a preregistered, multi-reviewer study using real tracking outputs, repeated hidden pairs, and a held-out judge-calibration set.

The diagnostic uses ground-truth poses. Some deployment videos have no reference trajectory; those require reference-free views and a separate claim. The fixed rubric may not represent every use case. Virtual production, robot navigation, and 3D reconstruction place different costs on short jumps, slow drift, and rotation error. Reporting component judgments is safer than presenting one universal score.

The Bayesian experiment uses a smooth synthetic objective. Gaussian processes can struggle in high-dimensional or conditional spaces. Search can also overfit the evaluator, which is why a held-out final set is required. Finally, the current implementation has not measured inter-rater reliability, learning effects, or vision-language judge calibration on a real comparison set.

8 Conclusion

Camera tracking needs more than one scalar, but informal visual inspection is not a reproducible replacement. Plumbline Reviewmakes that inspection testable: reject invalid output, render the same evidence, blind and balance comparisons, fit a preference model, resample by task, and keep the component metrics. Controlled perturbations show why the broader view matters, and the design-search experiment shows that the resulting expensive objective can be optimized under a budget. The method is ready for the next decisive test: blinded human comparisons of real camera tracks.

Data and Code Availability

Source code, schemas, tests, experiment scripts, generated result JSON, and paper figures are available at `https://github.com/kmatzen/plumbline`. Public model and dataset artifacts remain subject to their original licenses.

References

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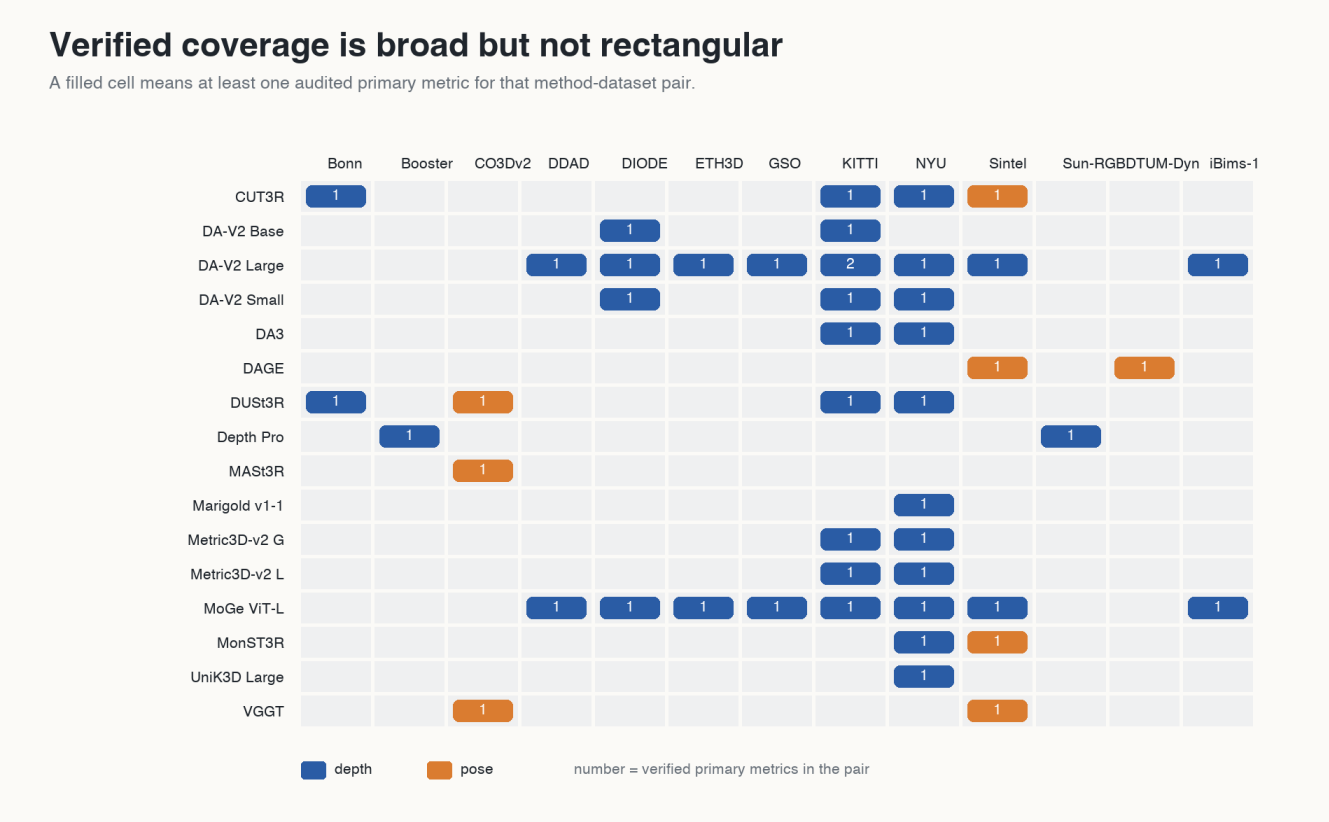


Figure 2: Supporting scope of the reproduction harness. Filled cells are verified published-result reproductions. Blank cells are untested, not failures. The new pairwise evaluator targets the smaller pose subset and future tracking integrations.

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